SET-F

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B.SC.[ECS] - II Sem III Practical Examination, October 2026

VSC1: Based on DSC1-3 and DSC1-4

Date & Time: Max Marks: 30

Instructions:

I) Attempt any **one** question from each section.

II) Each question carries **10 marks**.

III) The **Journal** and **Viva-Voce** carry **5 marks**.

Section-I: Based on DSC1-3 (Data Structures using C++)

Q.1 Write a program to search for an element in an unsorted array using linear search.

Q.2 Write a program to perform insert first, insert between, display operations of doubly linear linked list.

Section-II: Based on DSC1-4 (Database Management Systems)

Q.3 Create a table by applying proper constraints, insert at least 5 records and solve the following queries:

Field Name	Datatype	Size	Description
Sno	Varchar2	6	Primary Key
sname	Varchar2	20	Account Holders name
address	Varchar2	30	Residential address
address1	Varchar2	30	Permanent address
pincode	number	6	Pin code
Dob	Date		Date of birth
State	char	20	State
Doj	date		Date of joining must be greater than Date of birth
department	Char	20	department name
salary	number	9,2	Salary of salesman

1. Display the state which contains 5 characters.
2. Display all the salesmen whose salary is either 5000 or 10000.
3. Display all salesmen whose s_name contains the third letter 'n'.
4. Delete the record whose pincode is NULL.
5. Remove all the constraints from salesmen.

Q.4 Create a table by applying proper constraints, insert at least 5 records and solve the following queries:

A] Table Structure of Account_Master

Field Name	Datatype	Size	Description
Accnt	Varchar2	6	Primary Key
Acc_name	Varchar2	6	Account holder's name
Gender	Char	1	Either M or F
Op_Date	Date		The date on which the Account is opened
Chq_fac	Char	1	Cheque facility status set to default 'N'
Cl_Date	Date		

B] Table Structure of Account_Transaction

Field Name	Datatype	Size	Description
Accnt	Varchar2	6	Foreign Key
Tr_amount	Number	7,2	Transaction Amount
Tr_date	Date	1	Transaction Date
Chq_no	Varchar2	10	Cheque Number
Tr_type	Char	1	Receipt or Withdrawal

1. Display the details of the 2^{nd} maximum balance of the account holder.
2. Display the account_no, account_nm, transaction amount, transaction_type of all account holders.
3. Display the details of account holders who have the minimum balance.
4. Display the details of account holders whose month of opening the account and transaction are not the same.
5. Display the details of account holders who have not made any transactions and whose name's third letter is 'i'.